Expected Length of the Longest Common Subsequence

PETER RABINOVITCH
PETER.RABINOVITCH@GMAIL.COM

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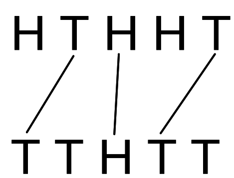
ABSTRACT. In this paper we review existing results on the longest common subsequence problem, also known as determining the Chvátal-Sankoff constant.

1. INTRODUCTION

You and I each toss a fair coin n times. What is the expected value of length of the longest common subsequence of the two sequences of coin tosses? The following example shows one of several longest common subsequences of length three.



Coin tossing problems, although simple to state, are at the core of probability, and can yield surprisingly rich phenomena, see [13] for example.

Although this problem can be easily explained, determining an exact expression for the expected value of the length of a longest common subsequence is still an unsolved problem after 30 years [4]. Various related problems have been proposed and solved, for example with large alphabets (e.g. using many sided dice, rather than coins [9]), and adding more independence to the problem [18], but so far, for the coin tossing problem, all we have are loose bounds [14] and simulation results [3].

Beyond the intellectual interest of the problem, it has application to determining when two strings are similar enough to be able to state that they are not significantly different, for example in computer security when extracting possible signatures for network worms [11], and in the biological context, with an alphabet of size 4, when comparing two sequences of DNA [22].

In this paper we review existing results on this and several related problems. The results are not new, although a few details not in the literature are included. In the next section, we formally define the problem, and state several preliminary results. Then we describe several different approaches to the problem, in the hope that the analysis of one of these approaches may yield information about the value of the Chvátal-Sankoff constant. We then report existing results for this problem, and follow with results for several closely related problems.

2. PROBLEM

In this section we formally state the problem, and provide preliminary observations.

Let $\mathbf{X} = \{\mathbf{X}_i\}_{1\leq i\leq m}$ and $\mathbf{Y} = \{\mathbf{Y}_j\}_{1\leq j\leq n}$ be two sequences of independent, identically distributed Bernoulli random variables, with $\mathbb{P}[\mathbf{X}_i = H] = \mathbb{P}[\mathbf{X}_i = T] = \mathbb{P}[\mathbf{Y}_j = H] =$

$\mathbb{P}[\mathbf{Y}_j = T] = \frac{1}{2}$. A common subsequence is a sequence that is a subsequence of both $\mathbf{X}$ and $\mathbf{Y}$. We are interested in the length of a *longest* common subsequence (there may be many), which we denote by $\mathbf{L}_{m,n}(\mathbf{X}, \mathbf{Y})$, and since $\mathbf{L}_{m,n}$ is a random variable, our target is $\mathbb{E}[\mathbf{L}_{m,n}]$. Even more simply, we are interested in the case where $m = n$, which we denote by $\mathbb{E}[\mathbf{L}_n]$. For small values of n, $\mathbb{E}[\mathbf{L}_n]$ can be computed, but for large values of n, it becomes computationally difficult to calculate $\mathbb{E}[\mathbf{L}_n]$ explicitly due to the large number of terms that must be evaluated. However, it is known that $\lim_{n\to\infty} \frac{1}{n}\mathbb{E}[\mathbf{L}_n] = c$ exists, and through simulations it is known that $c \simeq 0.81$. Determining the exact value of c, known as the Chvátal-Sankoff constant [4], is the goal of this research.

Let us calculate the means and variances of $\mathbf{L}_2$ and $\mathbf{L}_3$ by listing all possible choices for both sequences of tosses. In the following tables, we list the two sequences of coin tosses (by ordering of the sequences as if we replaced H by 1 and T by 0), a longest common subsequence of the two sequences of coin tosses (there may be several, we choose one arbitrarily), which we denote by LCS, and the length of a longest common subsequence of the two sequences of coin tosses, which we denote by L.

X	Y	$LCS(X,Y)$	$L(X,Y)$
TT	TT	TT	2
TT	TH	T	1
TT	HT	T	1
TT	HH	$\emptyset$	0
TH	TT	T	1
TH	TH	TH	2
TH	HT	H	1
TH	HH	H	1
HT	TT	T	1
HT	TH	H	1
HT	HT	HT	2
HT	HH	H	1
HH	TT	$\emptyset$	0
HH	TH	H	1
HH	HT	H	1
HH	HH	HH	2

Therefore,

$$\begin{aligned}
\mathbb{E}[\mathbf{L}_2] &= \frac{1}{16}(2 \times 0 + 10 \times 1 + 4 \times 2) \\
&= \frac{9}{8} \\
\mathbb{E}\left[\mathbf{L}_2^2\right] &= \frac{1}{16}\left(2 \times 0^2 + 10 \times 1^2 + 4 \times 2^2\right) \\
&= \frac{13}{8} \\
\mathbb{V}[\mathbf{L}_2] &= \frac{11}{8}
\end{aligned}$$

Similarly, for $n = 3$, we obtain

X	Y	$LCS(X,Y)$	$L(X,Y)$
TTT	TTT	TTT	3
TTT	TTH	TT	2
TTT	THT	TT	2
TTT	THH	T	1
TTT	HTT	TT	2
TTT	HTH	T	1
TTT	HHT	T	1
TTT	HHH	$\emptyset$	0
TTH	TTT	TT	2
TTH	TTH	TTH	3
TTH	THT	TH	2
TTH	THH	TH	2
TTH	HTT	TT	2
TTH	HTH	TH	2
TTH	HHT	H	1
TTH	HHH	H	1

X	Y	$LCS(X,Y)$	$L(X,Y)$
THT	TTT	TT	2
THT	TTH	TT	2
THT	THT	THT	3
THT	THH	TH	2
THT	HTT	TT	2
THT	HTH	TH	2
THT	HHT	HT	2
THT	HHH	H	1
THH	TTT	T	1
THH	TTH	TH	2
THH	THT	TH	2
THH	THH	THH	3
THH	HTT	T	1
THH	HTH	TH	2
THH	HHT	HH	2
THH	HHH	HH	2

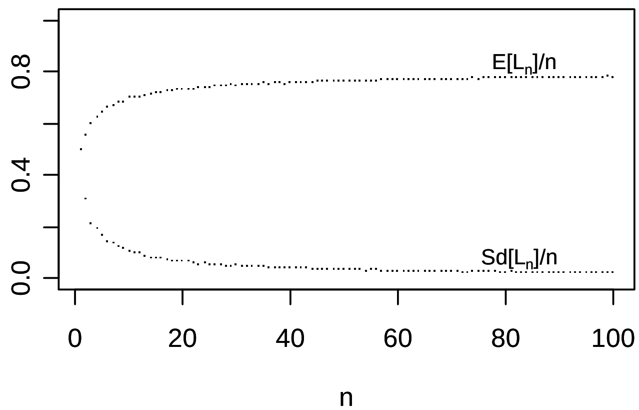
X	Y	$LCS(X,Y)$	$L(X,Y)$
HTT	TTT	TT	2
HTT	TTH	TT	2
HTT	THT	TT	2
HTT	THH	T	1
HTT	HTT	HTT	3
HTT	HTH	HT	2
HTT	HHT	HT	2
HTT	HHH	H	1
HTH	TTT	T	1
HTH	TTH	TH	2
HTH	THT	HT	2
HTH	THH	TH	2
HTH	HTT	HT	2
HTH	HTH	HTH	3
HTH	HHT	HH	2
HTH	HHH	HH	2

X	Y	$LCS(X,Y)$	$L(X,Y)$
HHT	TTT	T	1
HHT	TTH	T	1
HHT	THT	HT	2
HHT	THH	HH	2
HHT	HTT	HT	2
HHT	HTH	HH	2
HHT	HHT	HHT	3
HHT	HHH	HH	2
HHH	TTT	$\emptyset$	0
HHH	TTH	H	1
HHH	THT	H	1
HHH	THH	HH	2
HHH	HTT	H	1
HHH	HTH	HH	2
HHH	HHT	HH	2
HHH	HHH	HHH	3

Therefore,

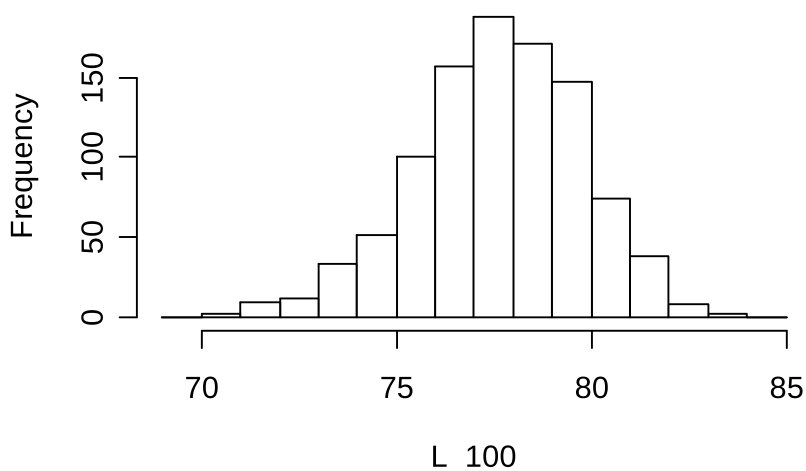
$$\begin{aligned}
\mathbb{E}[\mathbf{L}_3] &= \frac{1}{64}(2\times 0 + 16\times 1 + 38\times 2 + 8\times 3) \\
&= \frac{29}{16} \\
\mathbb{E}\left[\mathbf{L}_3^2\right] &= \frac{1}{64}\left(2\times 0^2 + 16\times 1^2 + 38\times 2^2 + 8\times 3^2\right) \\
&= \frac{15}{4} \\
\mathbb{V}[\mathbf{L}_3] &= \frac{119}{256}
\end{aligned}$$

It is interesting, and useful, to plot simulated values of $\mathbb{E}[\mathbf{L}_n]/n$ and the standard deviation of $\mathbf{L}_n/n$, based on 1000 simulations for each value of n:



It appears that $\mathbb{E}[\mathbf{L}_n]/n$ is monotone increasing, and that the standard deviation of $\mathbf{L}_n/n$ is monotone decreasing.

The following is a histogram of $\mathbf{L}_{100}$ based on 1000 simulation runs



Histogram of L_100 based on 1000 simulations of L_100.

Thus $\mathbf{L}_n$ appears to be approximately Gaussian, although other distributions can not be ruled out. To the best of the author's knowledge, there is no research on this issue.

However, in the related problem of the longest increasing subsequence in the plane (to be discussed later), it has been shown that the limiting distribution is the Tracy-Widom distribution [15].

We know that $\mathbb{E}\left[\mathbf{L}_n\right]/n \leq 1$, as a subsequence is not longer than the original sequence, and $\mathbb{E}\left[\mathbf{L}_1\right] = \frac{1}{2}$.

3. Approaches

In this section we look at different ways of determining the length of a longest common subsequence of two given sequences. It is our hope that insight obtained from these algorithms will allow for understanding what happens for two random sequences.

For a sequence X we use the notation $X_{i\ldots j}$ to denote the subsequence consisting of elements i through j (inclusive) of X where $i \leq j$. As a special case, we have $X_{i\ldots i} = X_i$, and for $i > j$ $X_{i\ldots j} = \emptyset$.

3.1. Dynamic Programming. In this section, we explain the dynamic programming approach to the longest common subsequence problem [23], and it is this approach that most computer programs actually use.

Dynamic programming is a method of solving problems that allow for merging of two smaller problems into the solution of a larger one [10].

Given two sequences X of length m and Y of length n we have the following

$$LCS\left(X_{1\ldots i}, Y_{1\ldots j}\right) = \begin{cases} \emptyset & \text{if } i = 0 \text{ or } j = 0 \\ LCS\left(X_{1\ldots i-1}, Y_{1\ldots j-1}\right) \& X_i & \text{if } X_i = Y_j \\ max\left\{LCS\left(X_{1\ldots i}, Y_{1\ldots j-1}\right), LCS\left(X_{1\ldots i-1}, Y_{1\ldots j}\right)\right\} & \text{otherwise} \end{cases}$$

Here, & denotes concatenation of sequences, and the *max* returns the longest sequence.

An example helps illustrate:Suppose $X = (H, T, H, H, T)$ and $Y = (H, H, T, H, T)$.

$$\begin{aligned}
LCS\left(X_{1\ldots 5}, Y_{1\ldots 5}\right) &= LCS\left(X_{1\ldots 4}, Y_{1\ldots 4}\right) + T \\
LCS\left(X_{1\ldots 4}, Y_{1\ldots 4}\right) &= LCS\left(X_{1\ldots 3}, Y_{1\ldots}\right) + H \\
LCS\left(X_{1\ldots 3}, Y_{1\ldots 3}\right) &= max\left\{LCS\left(X_{1\ldots 3}, Y_{1\ldots 2}\right), LCS\left(X_{1\ldots 2}, Y_{1\ldots 3}\right)\right\} \\
LCS\left(X_{1\ldots 3}, Y_{1\ldots 2}\right) &= LCS\left(X_{1\ldots 2}, Y_{1\ldots 1}\right) + H \\
LCS\left(X_{1\ldots 2}, Y_{1\ldots 1}\right) &= max\left\{LCS\left(X_{1\ldots 2}, Y_{1\ldots 0}\right), LCS\left(X_{1\ldots 1}, Y_{1\ldots 1}\right)\right\} \\
LCS\left(X_{1\ldots 2}, Y_{1\ldots 0}\right) &= \emptyset \\
LCS\left(X_{1\ldots 1}, Y_{1\ldots 1}\right) &= H \\
LCS\left(X_{1\ldots 2}, Y_{1\ldots 3}\right) &= LCS\left(X_{1\ldots 1}, Y_{1\ldots 2}\right) + T \\
LCS\left(X_{1\ldots 1}, Y_{1\ldots 2}\right) &= H
\end{aligned}$$

Thus

$$\begin{aligned}
LCS\left(X_{1\ldots 1}, Y_{1\ldots 2}\right) &= H \\
LCS\left(X_{1\ldots 2}, Y_{1\ldots 3}\right) &= HT \\
LCS\left(X_{1\ldots 2}, Y_{1\ldots 1}\right) &= H \\
LCS\left(X_{1\ldots 3}, Y_{1\ldots 2}\right) &= HH \\
LCS\left(X_{1\ldots 3}, Y_{1\ldots 3}\right) &= max\left\{HH, HT\right\} \\
LCS\left(X_{1\ldots 4}, Y_{1\ldots 4}\right) &= max\left\{HHH, HTH\right\} \\
LCS\left(X_{1\ldots 5}, Y_{1\ldots 5}\right) &= max\left\{HHHT, HTHT\right\}
\end{aligned}$$

and so there are two distinct longest common subsequences $(HHHT, HTHT)$ of length 4.

3.2. Patience Sorting. First we explain patience sorting, used in [1] to explain the length of a longest increasing subsequence of a uniformly random permutation, via an example, and then we explain the connection between patience sorting and the longest common subsequence.

Consider a list of distinct numbers $(3, 5, 2, 1, 7, 8, 9, 4, 6)$ which we will write down in a table. The first element of the list starts the first column of the table. We now proceed as follows: take the next element of the list, and compare it to the numbers at the bottom of the columns of the table, scanning from left to right. Put this element of the list at the bottom of the first column that the element is less than or equal to. If the element is larger than the bottom number of each column, start a new column with this element at the top.

$$\boxed{3} \rightarrow \boxed{\begin{array}{ll} 3 & 5 \end{array}} \rightarrow \boxed{\begin{array}{ll} 3 & 5 \\ 2 & \end{array}} \rightarrow \boxed{\begin{array}{ll} 3 & 5 \\ 2 & \\ 1 & \end{array}} \rightarrow \boxed{\begin{array}{lll} 3 & 5 & 7 \\ 2 & & \\ 1 & & \end{array}} \rightarrow$$

$$\boxed{\begin{array}{llll} 3 & 5 & 7 & 8 \\ 2 & & & \\ 1 & & & \end{array}} \rightarrow \boxed{\begin{array}{lllll} 3 & 5 & 7 & 8 & 9 \\ 2 & & & & \\ 1 & & & & \end{array}} \rightarrow \boxed{\begin{array}{lllll} 3 & 5 & 7 & 8 & 9 \\ 2 & 4 & & & \\ 1 & & & & \end{array}} \rightarrow \boxed{\begin{array}{lllll} 3 & 5 & 7 & 8 & 9 \\ 2 & 4 & 6 & & \\ 1 & & & & \end{array}}$$

From the results presented in [1], we see that a longest increasing subsequence of $(3, 5, 2, 1, 7, 8, 9, 4, 6)$ is $(3, 5, 7, 8, 9)$ of length 5. The subsequence is read off the top row of the table, and the number of columns in the table gives the length of a longest increasing subsequence.

Let us explain the connection between the longest common subsequence problem and patience sorting by an example. Let $X = (HTH)$ and $Y = (THH)$. Form the following lists: $y_T = (0)$ and $y_H = (1, 2)$ where y_i consists of the locations of i in Y, with indexing starting from 0. Then sort each of y_T and y_H in reverse order (from biggest to smallest) to get $\widetilde{y_T} = (0)$ and $\widetilde{y_H} = (2, 1)$. Now form a list $z = (2, 1, 0, 2, 1)$ obtained by replacing the i^{th} element of X with $\widetilde{y_T} = (0)$ and $\widetilde{y_H} = (2, 1)$ according to the value of the i^{th} element of X, i.e. if the i^{th} element of X is T, substitute $\widetilde{y_T}$, and if the i^{th} element of X is H, substitute $\widetilde{y_H}$.

Now perform patience sorting on z to obtain the following:

$$\begin{array}{ll} 2 & 2 \\ 1 & 1 \\ 0 & \end{array}$$

The number of columns in this table is the length of the longest common subsequence of X and Y.

The difference between the longest common subsequence problem, and the longest increasing subsequence problem is that our list, z, is not a permutation (although these techniques have been adapted to generalized permutations, in which elements may be repeated [21]). Also, when generating the list z from sequences of coin tosses,there is no reason to believe that z is distributed uniformly. Thus, it is not clear if the existing work on the longest increasing subsequence of a permutation can be generalized in a direction that can be useful to solve the longest common subsequence problem.

3.3. Last Passage Percolation. In this section, we connect the longest common subsequence problem to last passage percolation.

In a simple model of last passage percolation [16], we consider points on a two dimensional lattice, and at each vertex we assign a 'weight' chosen independently from some distribution. The weight of a path is defined to be the sum of the weights of the points the path passes through. In this model, we consider only paths that are strictly up-right, i.e. each coordinate is strictly increasing. We then ask what is the maximum weight of any path.

The longest common subsequence problem maps to a last passage percolation problem on a grid where each vertex is coloured red or green. Paths on the grid are allowed to use only green vertices. The choices of red and green points are not independent, but rather dependent on the sequences of coin tosses. Again, an example clarifies: Suppose $X = (H, T, H, H, T)$ and $Y = (H, H, T, H, T)$. We then form the following grid, putting X on the bottom, and Y going up the left side:

T					
H					
T					
H					
H					
	H	T	H	H	T

Next, we label each square of the grid red or green according to whether the coin tosses on the coordinate axes agree to get

T	red	green	red	red	green
H	green	red	green	green	red
T	red	green	red	red	green
H	green	red	green	green	red
H	green	red	green	green	red
	H	T	H	H	T

and in the longest common subsequence problem, we now seek the longest strictly up-right path on this grid that lands only on green squares. For example, in the following the bold squares are a longest common subsequence:

T	red	green	red	red	**green**
H	green	red	**green**	green	red
T	red	**green**	red	red	green
H	green	red	green	green	red
H	**green**	red	green	green	red
	H	T	H	H	T

and thus a longest common subsequence is $(HTHT)$ of length 4. The direct connection to the last passage percolation problem is seen by assigning weight 1 to green squares, and $-\infty$ to red squares.

Note that although the tosses are independent, the labeling of the squares of the grid is not, as there are $2n$ tosses and n^2 squares of the grid. For example, the second column of the table is entirely determined by the first column, and any single cell of the second column.

4. Key Results

In this section we first derive several key known results about this problem.

We first prove that the limit of $\frac{\mathbb{E}[\mathbf{L}_n]}{n}$, as $n \to \infty$ exists.

Definition 1. ***Subadditive*** *A sequence $\{a_n\}_{n\geq 1}$ is subadditive if $a_{m+n} \leq a_m + a_n$ for all positive integers m and n.*

Definition 2. ***Superadditive***. *A sequence $\{a_n\}_{n\geq 1}$ is superadditive if $\{-a_n\}_{n\geq 1}$ is subadditive, i.e. if $a_{m+n} \geq a_m + a_n$ for all positive integers m and n.*

Theorem 3. ***Fekete's Lemma***: *If $\{a_n\}_{n\geq 1}$ is subadditive, then*

$$\lim_{n\to\infty} \frac{a_n}{n} = \inf_n \frac{a_n}{n} \triangleq \gamma$$

where $\gamma \in [-\infty, \infty)$

Proof. [12] Consider first $\gamma > -\infty$. For any $\varepsilon > 0$ we can find a k such that $a_k \leq (\gamma + \varepsilon) k$ because $\gamma = \inf_n \frac{a_n}{n}$. Since any $m > 0$ can be written as $m = nk + j$ with $0 \leq j < k$ for the same k, it follows that

$$\begin{aligned} a_m &= a_{nk+j} \leq a_{nk} + a_j \\ &\leq na_k + a_j \\ &\leq (\gamma + \varepsilon) nk + \max_{0\leq l<k} a_l \end{aligned}$$

so

$$\begin{aligned} \frac{a_m}{m} &\leq \frac{(\gamma + \varepsilon) nk}{m} + \frac{\max_{0\leq l<k} a_l}{m} \\ \limsup_m \frac{a_m}{m} &\leq \limsup_m \frac{(\gamma + \varepsilon) nk}{m} + \limsup_m \frac{\max_{0\leq l<k} a_l}{m} \\ \limsup_m \frac{a_m}{m} &\leq \gamma + \varepsilon \end{aligned}$$

and

$$\gamma + \varepsilon \leq \lim \inf_{n\to\infty} \frac{a_n}{n} + \varepsilon$$

so

$$\limsup_m \frac{a_m}{m} \leq \gamma + \varepsilon \leq \liminf_m \frac{a_m}{m} + \varepsilon$$

and thus, since $\varepsilon > 0$ was arbitrary, we have

$$\lim_{n\to\infty} \frac{a_n}{n} = \inf_n \frac{a_n}{n} = \gamma.$$

The case of $\gamma = -\infty$ follows since then we would have (from above) that $\limsup_m \frac{a_m}{m} \leq \gamma + \varepsilon = -\infty$, and hence $\lim_{n\to\infty} \frac{a_n}{n} = -\infty$. ■

Corollary 4. *If $\{a_n\}_{n\geq 1}$ is superadditive, then*

$$\lim_{n\to\infty} \frac{a_n}{n} = \sup_n \frac{a_n}{n} = \gamma$$

where $\gamma \in (-\infty, \infty]$.

Proof. Apply Fekete's Lemma to $\{-a_n\}_{n\geq 1}$. ■

Theorem 5. $a_n \triangleq \mathbb{E}[\mathbf{L}_n]$ *is superadditive*

Proof. $\mathbb{E}[\mathbf{L}_{r+s}] \geq \mathbb{E}[\mathbf{L}_r] + \mathbb{E}[\mathbf{L}_s]$ by concatenating a longest common subsequence from a block of r pairs of coin tosses with a longest common subsequence from a block of s pairs of coin tosses. ■

Corollary 6. $\lim_{n\to\infty} \frac{\mathbb{E}[\mathbf{L}_n]}{n}$ *exists.*

Now we show that $\mathbb{P}\left[|\mathbf{L}_n - \mathbb{E}[\mathbf{L}_n]| \geq \lambda\sqrt{n}\right] \leq 2e^{-\frac{\lambda^2}{8}}$.

Theorem 7. ***Azuma Hoeffding Inequality*** *[12]. Suppose a sequence of random variables $\mathbf{X}_n$ forms a martingale with respect to a σ-field $\mathcal{F}_n$, and that $\mathbf{X}_0 = 0$. If there exists a sequence of constants c_n such that $\mathbb{P}[|\mathbf{X}_n - \mathbf{X}_{n-1}| \leq c_n] = 1$, then*

$$\mathbb{E}\left[e^{\beta \mathbf{X}_n}\right] \leq e^{\frac{\beta^2}{2}\sum_{k=1}^n c_k^2}$$

for all $\beta > 0$. Furthermore

$$\mathbb{P}[\mathbf{X}_n \geq \lambda] \leq e^{-\lambda^2/\left(2\sum_{k=1}^n c_k^2\right)}$$

and

$$\mathbb{P}[|\mathbf{X}_n| \geq \lambda] \leq 2e^{-\lambda^2/\left(2\sum_{k=1}^n c_k^2\right)}$$

for all $\lambda > 0$.

Proof. Because the function *exp* is convex, $\exp(\alpha u + (1-\alpha)v) \leq \alpha \exp(u) + (1-\alpha)\exp(v)$ for any $\alpha \in [0,1]$, u and v. Setting $u = -\beta c$, $v = \beta c$ and $\alpha = \frac{c-x}{2c}$ for $x \in [-c, c]$ yields

$$e^{\beta x} \leq \frac{c-x}{2c}e^{-\beta c} + \frac{c+x}{2c}e^{\beta c}.$$

If we substitute $\mathbf{X}_n - \mathbf{X}_{n-1}$ for x and c_n for c in this expression and take conditional expectations, then the hypothesis $|\mathbf{X}_n - \mathbf{X}_{n-1}| \leq c_n$ and the fact that $\mathbb{E}[\mathbf{X}_n - \mathbf{X}_{n-1}|\mathcal{F}_{n-1}] = 0$ imply

$$\begin{aligned}
\mathbb{E}\left[e^{\beta\mathbf{X}_n}\right] &= \mathbb{E}\left[\mathbb{E}\left[e^{\beta\mathbf{X}_n}|\mathcal{F}_{n-1}\right]\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[e^{\beta\mathbf{X}_{n-1}+\beta\mathbf{X}_n-\beta\mathbf{X}_{n-1}}|\mathcal{F}_{n-1}\right]\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[e^{\beta\mathbf{X}_{n-1}}e^{\beta(\mathbf{X}_n-\mathbf{X}_{n-1})}|\mathcal{F}_{n-1}\right]\right] \\
&= \mathbb{E}\left[e^{\beta\mathbf{X}_{n-1}}\mathbb{E}\left[e^{\beta(\mathbf{X}_n-\mathbf{X}_{n-1})}|\mathcal{F}_{n-1}\right]\right] \\
&\leq \mathbb{E}\left[e^{\beta\mathbf{X}_{n-1}}\mathbb{E}\left[\frac{c_n-(\mathbf{X}_n-\mathbf{X}_{n-1})}{2c_n}e^{-\beta c_n} + \frac{c_n+(\mathbf{X}_n-\mathbf{X}_{n-1})}{2c_n}e^{\beta c_n}|\mathcal{F}_{n-1}\right]\right] \\
&= \mathbb{E}\left[e^{\beta\mathbf{X}_{n-1}}\left[\frac{c_n-\mathbb{E}[\mathbf{X}_n-\mathbf{X}_{n-1}|\mathcal{F}_{n-1}]}{2c_n}e^{-\beta c_n} + \frac{c_n+\mathbb{E}[\mathbf{X}_n-\mathbf{X}_{n-1}|\mathcal{F}_{n-1}]}{2c_n}e^{\beta c_n}\right]\right] \\
&= \mathbb{E}\left[e^{\beta\mathbf{X}_{n-1}}\left(\frac{c_n-0}{2c_n}e^{-\beta c_n} + \frac{c_n+0}{2c_n}e^{\beta c_n}\right)\right] \\
&= \mathbb{E}\left[e^{\beta\mathbf{X}_{n-1}}\right]\left(\frac{c_n}{2c_n}e^{-\beta c_n} + \frac{c_n}{2c_n}e^{\beta c_n}\right) \\
&= \mathbb{E}\left[e^{\beta\mathbf{X}_{n-1}}\right]\left(\frac{e^{-\beta c_n}+e^{\beta c_n}}{2}\right)
\end{aligned}$$

where the third to last line is because $\mathbf{X}_n$ is a martingale, and $\mathbf{X}_0 = 0$. Induction on n now verifies the first part of the theorem, provided we can show that

$$\frac{e^u + e^{-u}}{2} \le e^{\frac{u^2}{2}}.$$

However, this follows by expanding both sides in a Taylor series

$$\begin{aligned} \frac{e^u + e^{-u}}{2} &= \frac{1}{2}\sum_{n=0}^{\infty} \frac{u^n + (-u)^n}{n!} \\ e^{\frac{u^2}{2}} &= \sum_{n=0}^{\infty} \left(\frac{u^{2n}}{2^n n!}\right) \end{aligned}$$

and noting that the corresponding coefficients of u^{2n} satisfy $\frac{1}{(2n)!} \le \frac{1}{2^n n!}$, as can be verified by induction. Also, the coefficients of the odd powers u^{2n+1} on both sides vanish.

To prove the second part of the theorem, we apply Markov's inequality in the form

$$\begin{aligned} \mathbb{P}\left[\mathbf{X}_n \ge \lambda\right] &\le \mathbb{E}\left[e^{\beta \mathbf{X}_n}\right] e^{-\beta\lambda} \\ &\le e^{\frac{\beta^2}{2}\sum_{k=1}^n c_k^2 - \beta\lambda} \end{aligned}$$

for arbitrary $\beta > 0$. The particular β that minimizes the exponent on the right hand side is clearly $\beta = \lambda / \sum_{k=1}^n c_k^2$. Substituting this choice into the above yields the result. Because $-\mathbf{X}_n$ also satisfies the hypothesis of the theorem, the final part of the theorem follows form the second part. ■

Theorem 8. *[12]* $\mathbb{P}\left[|\mathbf{L}_n - \mathbb{E}\left[\mathbf{L}_n\right]| \ge \lambda\sqrt{n}\right] \le 2e^{-\frac{\lambda^2}{8}}$

Proof. Let $\mathbf{Y}_i$ be the random pair of letters chosen for position i of the two strings. Conditioning on the σ-field $\mathcal{F}_i$ generated by $\mathbf{Y}_1, \mathbf{Y}_2, \ldots \mathbf{Y}_i$ creates a martingale $\mathbf{X}_i = \mathbb{E}\left[\mathbf{L}_n|\mathcal{F}_i\right]$ with $\mathbf{X}_i = \mathbf{L}_i$. We now bound $|\mathbf{X}_i - \mathbf{X}_{i-1}|$ by considering what happens when exactly one argument of $\mathbf{L}_n(y_1, y_2, \ldots, y_n)$ changes. If this is argument i, then the pair $y_i = (u_i, v_i)$ becomes $y_i^* = (u_i^*, v_i^*)$. In a longest common subsequence of $y = (y_1, y_2, \ldots, y_n)$, changing u_i to u_i^* creates or destroys at most one match, and similarly for the v_i. Thus, the new string y^* gas a common subsequence whose length differs from $\mathbf{L}_n(y)$ by at most 2. The same argument shows that y has a common subsequence whose length differs from $\mathbf{L}_n(y^*)$ by at most 2. Thus $|\mathbf{L}_n(y) - \mathbf{L}_n(y^*)| \le 2$, and thus

$$\mathbb{P}\left[|\mathbf{L}_n - \mathbb{E}\left[\mathbf{L}_n\right]| \ge \lambda\sqrt{n}\right] \le 2e^{-\frac{\lambda^2}{8}}$$

■

For the longest common subsequence problem with coin flips, [7] shows, using Kingman's subadditive ergodic theorem [8, Theorem 6.1], that $\frac{\mathbf{L}_n}{n}$ converges to a constant almost surely, and in [2], it is shown that there exists a constant C such that for all n

$$\delta n \ge \mathbb{E}\mathbf{L}_n \ge \delta n - C\sqrt{n\log n}.$$

for some $\delta \in [0,1]$.

In [14], it is shown, via building finite state machines that can be modeled by Markov chains, that

$$0.788071 \le \frac{\mathbb{E}\left[\mathbf{L}_n\right]}{n} \le 0.826280.$$

and in [3] it is shown via simulation that $\frac{\mathbb{E}[\mathbf{L}_n]}{n} = 0.812653$.

5. Related Problems

In this section, we present some of the more interesting results on closely related problems.

So far, we have been focussing on the case of coin flips, i.e. where there are two possible outcomes for every toss. It is also possible to consider the case where there are k possible outcomes. For $k = 4$, this has application to strings of DNA, $k = 20$ has applications to strings of proteins, and $k = 256$ has application to strings of bytes, for example in comparing computer files. Although no exact results exist for the general k case, recently in [9] it was shown that $\gamma_k\sqrt{k} \to 2$ as $k \to \infty$., where $\gamma_k = \lim_{n\to\infty} \mathbb{E}\left[\mathbf{L}_n(k)\right]$, with $\mathbf{L}_n(k)$ denoting the length of a longest common subsequence of n letters, chosen uniformly for an alphabet with k letters.

The most fully developed problem that is closely related is the so called 'Bernoulli model', where rather than having points on the grid labeled red or green by two sequences of coin flips, as above, each point on the grid is labeled red or green independently with probability $1/2$. Here, we denote the length of the limiting longest up-right path divided by the dimension of the grid, stepping only on green squares, by γ. For the Bernoulli model, it was shown in [18] that $\gamma = 2\sqrt{2} - 2$. It is intriguing that the same value was conjectured for c by Steele & Arratia [19], but later shown to disagree with results from simulation [3]. They argued heuristically as follows: call any pair of subsequences of length k where the $\mathbf{X}_i$ and $\mathbf{Y}_i$ agree a 'good k pair'. Let $\mathbf{Z}$ be the total number of good k pairs of the two length n strings X and Y. Then

$$\mathbb{E}\left[\mathbf{Z}\right] = \binom{n}{k}^2 \frac{1}{2^k}$$

because there are $\binom{n}{k}$ ways to choose each of the two subsequences, which have to agree in k places. The mode of this sequence is determined by looking at ratios of successive terms:

$$\begin{aligned}
a_k \triangleq \frac{\binom{n}{k+1}^2 \frac{1}{2^{k+1}}}{\binom{n}{k}^2 \frac{1}{2^k}} &= \frac{\binom{n}{k+1}^2}{\binom{n}{k}^2} \frac{2^k}{2^{k+1}} \\
&= \left[\frac{n!}{(k+1)!\,(n-(k+1))!} \frac{k!\,(n-k)!}{n!}\right]^2 \frac{1}{2} \\
&= \left(\frac{n-k}{k+1}\right)^2 \frac{1}{2}
\end{aligned}$$

Now a_k is greater than one when

$$\begin{aligned}
a_k &\geq 1 \\
\left(\frac{n-k}{k+1}\right)^2 \frac{1}{2} &\geq 1 \\
\left(\frac{n-k}{k+1}\right)^2 &\geq 2 \\
\left(\frac{n-k}{k+1}\right) &\geq \sqrt{2} \\
\frac{n}{k+1} - \frac{k}{k+1} &\geq \sqrt{2}
\end{aligned}$$

$$\begin{aligned}
\frac{n}{k+1} - \frac{k+1}{k+1} + \frac{1}{k+1} &\geq \sqrt{2} \\
\frac{n}{k+1} - 1 + \frac{1}{k+1} &\geq \sqrt{2} \\
\frac{n+1}{k+1} &\geq \sqrt{2}+1 \\
\frac{n+1}{\sqrt{2}+1} - 1 &\geq k
\end{aligned}$$

thus the mode of this sequence is approximately $\frac{n}{\sqrt{2}+1}$. Note also that since every length k subsequence of a longest common subsequence yields a good k pair, we see that there are at least $\binom{\mathbf{L}_n}{k}$ such good k pairs. This sequence has mode $\frac{\mathbf{L}_n}{2}$. By equating the two modes, we find that

$$\begin{aligned}
\frac{\mathbf{L}_n}{2} &= \frac{n}{\sqrt{2}+1} \\
\frac{\mathbf{L}_n}{n} &= \frac{2}{\sqrt{2}+1} \\
&= \frac{2}{\sqrt{2}+1}\frac{\sqrt{2}-1}{\sqrt{2}-1} \\
&= 2\sqrt{2}-2
\end{aligned}$$

Finally, it was shown recently [15], with $\mathbf{L}_n$ denoting the length of the longest strictly up-right path in the Bernoulli model on a grid of size n, that for large n

$$\mathbf{L}_n \approx \left(2\sqrt{2}-2\right)n + 2^{1/6}\left(\sqrt{2}-1\right)^{4/3} n^{1/3}\chi$$

where χ has the Tracy-Widom distribution [6].

6. CONCLUSION

It is both challenging and frustrating that a problem as simple to state as the longest common subsequence problem does not have a solution. The agreement between the Arratia-Steele conjectured value for c and the actual value for γ is intriguing, and given the closeness of these two constants, perhaps there is a way to compare the geometry of the grids in the Bernoulli and LCS models to bound the difference between the two constants.

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